THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 13: **Algebra Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A three-digit number with digits a, b, c is reversed. If $a - c = 4$, the difference between the number and its reverse is: 1

(a) 360 **(b)** 400 **(c)** 36 **(d)** 396

2. A two-digit number exceeds the number formed by reversing its digits by 54. The difference of its digits is: 1

(a) 5 **(b)** 6 **(c)** 7 **(d)** 54

3. If $x + \frac{1}{x} = 6$, then $x^2 + \frac{1}{x^2}$ equals: 1

(a) 38 **(b)** 12 **(c)** 36 **(d)** 34

4. The sum of the dates in a 2×2 calendar block is 32. The smallest date in the block is: 1

(a) 12 **(b)** 3 **(c)** 5 **(d)** 4

5. If $a^2 - b^2 = 345$ and $a - b = 15$, then $a + b$ equals: 1

(a) 15 **(b)** 76 **(c)** 24 **(d)** 23

6. The value of $46^2 - 43 \times 49$ is: 1

(a) 6 **(b)** 0 **(c)** 46 **(d)** 9

7. The sum of three consecutive integers is 201. The largest of them is: 1

(a) 69 **(b)** 67 **(c)** 68 **(d)** 66

8. A three-digit number with digits a, b, c is reversed. If $a - c = 5$, the difference between the number and its reverse is: 1

(a) 45 **(b)** 495 **(c)** 500 **(d)** 450

9. **Assertion (A):** The difference of the squares of two consecutive numbers $n + 1$ and n is $2n + 1$. 1
Reason (R): $a^2 - b^2 = (a + b)(a - b)$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** If the digits of a two-digit number are a (tens) and b (units), the number obtained by reversing the digits is $10b + a$. 1
Reason (R): The sum of a two-digit number and its reverse is $11(a + b)$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Prove that $(n+3)^2 - (n-3)^2 = 12n$ and use it to find the value of $(13+3)^2 - (13-3)^2$. 2

12. The sum of the digits of a two-digit number is 8 and the tens digit exceeds the ones digit by 4. Find the number. 2

13. If $x + y = 45$ and $x - y = 6$, find $x^2 - y^2$. 2

14. Prove that $(n+3)^2 - (n-3)^2 = 12n$ and use it to find the value of $(20+3)^2 - (20-3)^2$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The sum of a two-digit number and its reverse is 176 and the digits differ by 2. Find the number (tens digit larger). 3

16. If $x + \frac{1}{x} = 7$, find $x^2 + \frac{1}{x^2}$ and $x^4 + \frac{1}{x^4}$. 3

17. The sum of the dates of a 2×2 block in a calendar is 100. Find the four dates and show that the block's sum is always $4\times$(smallest date)$+16$. 3

OR

The sum of a two-digit number and its reverse is 99 and the digits differ by 3. Find the number (tens digit larger). 3

18. If $x + \frac{1}{x} = 8$, find $x^2 + \frac{1}{x^2}$ and $x^4 + \frac{1}{x^4}$. 3

OR

The sum of the dates of a 2×2 block in a calendar is 48. Find the four dates and show that the block's sum is always $4\times$(smallest date)$+16$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A teacher says: Take any two-digit number, add the number formed by reversing its digits, and the sum is always a multiple of 11. Let the tens digit be a and the units digit b. 5
(i) Write the number in terms of a and b.
(ii) Write the number formed by reversing the digits.
(iii) Find the sum of the two.
(iv) Hence show that the sum is a multiple of 11 and find it for 47.

20. Prove that $(6n+1)^2 - (6n-1)^2 = 24n$ and use it to evaluate it for $n = 2$. 5

OR

A three-digit number has digits a, b, c with $a - c = 6$. Show that the number minus its reverse is divisible by 99 and find the difference. 5

END OF QUESTION PAPER — Total: 40 Marks